

which has $2n$ distinct blocks of width $\geq 2^{k-1}$. The blocks are distinct since $3d - \mu_n > \dots > 3d - \mu_1 > d > \mu_1 > \dots > \mu_n$. By Lemma 4.6, the degree d' obtained after this sequence of $\ell - 1 = n2^{k-2}$ $*$ -seeds is at most

$$d' \leq \left(\frac{n2^{k-1} + 3}{3} \right) (5d) \leq 5n2^k d.$$

□

5.2. Lower bound on length increase. Here we prove the lower bound in Theorem 2.5, which we recall in the next proposition.

Proposition 5.2. *There exists a sequence of proper homaloidal types $(\mathbf{y}_N)_{N \geq 1}$ of degrees $d(\mathbf{y}_N) \leq 2^{N^2} 10^N$, such that $s(\mathbf{y}_N) \geq 2^{N-1}$. In particular, for any $c < \sqrt{\ln(2)}$ and N large enough,*

$$s(\mathbf{y}_N) \geq \exp(c\sqrt{\ln(d(\mathbf{y}_N)))}$$

Proof. Let $N \geq 1$ be an integer parameter. We begin by constructing recursively an auxiliary finite sequence of homaloidal types $(\mathbf{x}_{0,N}, \dots, \mathbf{x}_{N-1,N})$ whose last element will be $\mathbf{y}_N$. If we start with the de Jonquières homaloidal type

$$\mathbf{x}_{0,N} := (2^{N-1} + 1; 2^{N-1}, (1)^{2^N})$$

which has 1 block of width 2^N , we can apply Lemma 5.1 successively $N - 1$ times with $n = 2^i$ and $k = N - i$, where $i = 0, \dots, N - 2$. Doing so, we obtain a sequence of homaloidal types $(\mathbf{x}_{i,N}) = (\mathbf{x}_{0,N}, \dots, \mathbf{x}_{N-1,N})$ of degrees $d_{0,N} = 2^{N-1} + 1, d_{1,N}, \dots, d_{N-1,N}$ such that each $\mathbf{x}_{i,N}$ has 2^i distinct blocks of width $\geq 2^{N-i}$. The successive degrees satisfy the inequality

$$d_{i+1,N} \leq 5 \cdot 2^{N-i} \cdot 2^i \cdot d_{i,N} = 5 \cdot 2^N \cdot d_{i,N}.$$

At the end,

$$d_{N-1,N} \leq 5^N 2^{N^2} (2^{N-1} + 1) \tag{5.1}$$

$$\leq 2^{N^2} 10^N. \tag{5.2}$$

We now define the homaloidal type $\mathbf{y}_N := \mathbf{x}_{N-1,N}$, so $d(\mathbf{y}_N) \leq 2^{N^2} 10^N$. Its seedbed $s(\mathbf{y}_N)$ is larger than the number 2^{N-1} of blocks of width 2 that have been created. The last statement follows from the fact that when N is large, we have

$$N \geq \sqrt{\frac{\ln(d(\mathbf{y}_N))}{\ln(2)}} (1 + o(1)), \tag{5.3}$$